

and M2, or remove 5V/9V supply from J1 and then connect J2. Now, set the address code for the data (remote key) to be stored in the MCU using DIP switch DIP1 (using S2 through S5).

After setting the 4-bit address in hexadecimal code (say, 0101h), switch on the power using S1. After self-test, LED6 will glow, indicating the mode as recording. Press a button (say, 2) on the remote. LED9 will blink, indicating that a signal/

data is received by IRRX1 sensor. This signal is fed to the MCU, which processes and saves data in its in-built EEPROM.

At the same time, a visual indication of the address being assigned for each data is indicated by the glowing of corresponding LED2 through LED5 in equivalent hexadecimal format.

Data being saved is acknowledged by the blinking of LED7. This process can be repeated for up to 16 required codes (0 to F hex).

For easy control of the robot, we have assigned the code (or bits) to five buttons of the remote control as listed in Table 1.

For any other button, the code is set as stop (0000). So to stop the robot in an emergency case, press any other button on the remote.

After recording the required codes, remove J2 and press reset switch S6. After self-test, LED7 will glow, indicating that it is in play mode. Now, press the programmed

remote button randomly and check that LED2 through LED5 indicate the required code. LED8 will blink for a while if there are errors like a button not pressed properly or signal not received properly.

## Software

The software (remote\_robot.c) is written in C

## EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at [source.efymag.com](http://source.efymag.com)

## PARTS LIST

### Semiconductors:

- IC1 - 7805, 5V voltage regulator
- IC2 - ATmega8A MCU
- IC3 - L293D motor driver
- LED1-LED9 - 5mm LED
- D1 - 1N4007 rectifier diode

### Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1-R5 - 4.7-k $\Omega$ -ohm
- R6 - 10-k $\Omega$ -ohm
- R7 - 330-ohm
- R8-R16 - 1.5-k $\Omega$ -ohm

### Capacitors:

- C1, C2 - 4.7 $\mu$ F, 16V electrolytic
- C3-C5 - 0.1 $\mu$ F ceramic disk

### Miscellaneous:

- CON1 - 2-pin terminal connector
- CON2 - 3-pin connector
- CON3 - 2-pin connector
- CON4 - 6-pin berg strip male connector

- S1 - On/off switch
- S6 - Tactile switch
- DIP1 - 4-way DIP switch (for S2-S5 setting)
- IRRX1 - TSOP1738 receiver module or equivalent
- M1, M2 - 5V/9V DC motor
- J1, J2 - Shorting jumpers
- J3-J7 - Single-strand wire
- J8 - Small caster wheel
- J9 - Two small wheels for the motors
- J10 - 9V battery
- J11 - 28-pin IC base

| Button/key | Function   | Code |
|------------|------------|------|
| 2          | Forward    | 0101 |
| 4          | Left turn  | 0100 |
| 5          | Stop       | 0000 |
| 6          | Right turn | 0001 |
| 8          | Backward   | 1010 |

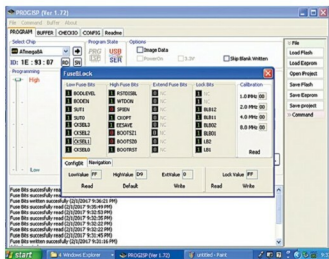


Fig. 2: Screenshot of fuse bit settings

### PCB Layout Service

- Schematic to Gerber Package
- Online Price Calculator
- No Exception during PCB Fabrication
- Your Virtual Design Team

### Stencil Fabrication

- Online Price Calculator
- With Frame & Frame-less
- Laser Cut Precision
- Delivery from 1 working day

B-24, GIDC Electronics Estate, Sector-25, Gandhinagar-382044, Gujarat, India | [pcb.pcbpower.com](http://pcb.pcbpower.com) | +91 7600012414 / +91 7600012415 | [www.pcbpower.com](http://www.pcbpower.com)

**20 YEARS OF EXCELLENCE**